

DBMS – PAYMENT SYSTEM MANAGEMENT

TITAN EYE – EYEWEAR E-COMMERCE DATABASE SYSTEM

1. DATABASE DESIGN

1.1 PAYMENT TABLE

Attribute	Description
Payment_ID	Unique payment ID
Order_ID	Related order ID
Payment_Date	Date of payment
Payment_Method	UPI / Card / Cash on Delivery / Net Banking
Payment_Amount	Amount paid
Payment_Status	Paid / Pending / Failed

2. TABLE CREATION

2.1 Create Payment Table

```
SQL> CREATE TABLE Payment (  
2     Payment_ID NUMBER PRIMARY KEY,  
3     Order_ID NUMBER,  
4     Payment_Date DATE,  
5     Payment_Method VARCHAR2(30),  
6     Payment_Amount NUMBER(10,2),  
7     Payment_Status VARCHAR2(30),  
8     FOREIGN KEY (Order_ID) REFERENCES Orders(Order_ID)  
9 );
```

Table created.

3. VALUE INSERTION

```

SQL> INSERT INTO Payment VALUES
  2 (501, 101, TO_DATE('20-09-2026','DD-MM-YYYY'), 'UPI', 2500.00, 'Paid');

1 row created.

SQL> INSERT INTO Payment VALUES
  2 (502, 102, TO_DATE('21-09-2026','DD-MM-YYYY'), 'Card', 1800.00, 'Paid');

1 row created.

SQL> INSERT INTO Payment VALUES
  2 (503, 103, TO_DATE('22-09-2026','DD-MM-YYYY'), 'Cash on Delivery', 3200.00, 'Pending');

1 row created.

SQL> INSERT INTO Payment VALUES
  2 (504, 104, TO_DATE('23-09-2026','DD-MM-YYYY'), 'Net Banking', 1500.00, 'Failed');

1 row created.

SQL> INSERT INTO Payment VALUES
  2 (505, 105, TO_DATE('24-09-2026','DD-MM-YYYY'), 'UPI', 2750.00, 'Paid');

1 row created.

SQL>
SQL> COMMIT;

Commit complete.

```

4. DISPLAY SUCCESSFUL PAYMENTS

```

SQL> SELECT *
  2 FROM Payment
  3 WHERE Payment_Status = 'Paid';

```

PAYMENT_ID	ORDER_ID	PAYMENT_D	PAYMENT_METHOD	PAYMENT_AMOUNT	PAYMENT_STATUS
501	101	20-SEP-26	UPI	2500	Paid
502	102	21-SEP-26	Card	1800	Paid
505	105	24-SEP-26	UPI	2750	Paid

5. DISPLAY FAILED / PENDING PAYMENTS

```

SQL> SELECT *
  2 FROM Payment
  3 WHERE Payment_Status IN ('Failed', 'Pending');

```

PAYMENT_ID	ORDER_ID	PAYMENT_D	PAYMENT_METHOD	PAYMENT_AMOUNT	PAYMENT_STATUS
503	103	22-SEP-26	Cash on Delivery	3200	Pending
504	104	23-SEP-26	Net Banking	1500	Failed

6. MODIFY PAYMENT STATUS

```
SQL> UPDATE Payment
  2 SET Payment_Status = 'Paid'
  3 WHERE Payment_ID = 503;
```

1 row updated.

```
SQL>
SQL> COMMIT;
```

Commit complete.

7. ANALYZE PAYMENT METHODS

7.1 COUNT TRANSACTIONS BY PAYMENT METHOD

```
SQL> SELECT Payment_Method, COUNT(*) AS Total_Transactions
  2 FROM Payment
  3 GROUP BY Payment_Method;
```

PAYMENT_METHOD	TOTAL_TRANSACTIONS
UPI	2
Card	1
Cash on Delivery	1
Net Banking	1

7.2 TOTAL AMOUNT BY PAYMENT METHOD

```
SQL> SELECT Payment_Method,
  2 SUM(Payment_Amount) AS Total_Amount
  3 FROM Payment
  4 GROUP BY Payment_Method;
```

PAYMENT_METHOD	TOTAL_AMOUNT
UPI	5250
Card	1800
Cash on Delivery	3200
Net Banking	1500

8. PAYMENT TRANSACTION REPORT

```
SQL> SELECT Payment_Method,
2      SUM(Payment_Amount) AS Total_Amount
3 FROM Payment
4 GROUP BY Payment_Method;
```

PAYMENT_METHOD	TOTAL_AMOUNT
UPI	5250
Card	1800
Cash on Delivery	3200
Net Banking	1500

```
SQL> SELECT
2      Payment_ID,
3      Order_ID,
4      Payment_Date,
5      Payment_Method,
6      Payment_Amount,
7      Payment_Status
8 FROM Payment
9 ORDER BY Payment_Date;
```

PAYMENT_ID	ORDER_ID	PAYMENT_D	PAYMENT_METHOD	PAYMENT_AMOUNT
501	101	20-SEP-26	UPI	2500
502	102	21-SEP-26	Card	1800
503	103	22-SEP-26	Cash on Delivery	3200

PAYMENT_ID	ORDER_ID	PAYMENT_D	PAYMENT_METHOD	PAYMENT_AMOUNT
504	104	23-SEP-26	Net Banking	1500
505	105	24-SEP-26	UPI	2750

9. CUSTOMER-WISE PAYMENT REPORT

```

SQL> SELECT
  2     o.Customer_ID,
  3     p.Order_ID,
  4     p.Payment_ID,
  5     p.Payment_Date,
  6     p.Payment_Method,
  7     p.Payment_Amount,
  8     p.Payment_Status
  9 FROM Orders o
 10 JOIN Payment p
 11 ON o.Order_ID = p.Order_ID
 12 ORDER BY o.Customer_ID;

```

CUSTOMER_ID	ORDER_ID	PAYMENT_ID	PAYMENT_D	PAYMENT_METHOD	PAYMENT_AMOUNT	PAYMENT_STATUS
1	101	501	20-SEP-26	UPI	2500	Paid
2	102	502	21-SEP-26	Card	1800	Paid
3	103	503	22-SEP-26	Cash on Delivery	3200	Paid

CUSTOMER_ID	ORDER_ID	PAYMENT_ID	PAYMENT_D	PAYMENT_METHOD	PAYMENT_AMOUNT	PAYMENT_STATUS
4	104	504	23-SEP-26	Net Banking	1500	Failed
5	105	505	24-SEP-26	UPI	2750	Paid

10.FINAL PAYMENT TABLE

```

SQL> SELECT * FROM Payment;

```

PAYMENT_ID	ORDER_ID	PAYMENT_D	PAYMENT_METHOD	PAYMENT_AMOUNT	PAYMENT_STATUS
501	101	20-SEP-26	UPI	2500	Paid
502	102	21-SEP-26	Card	1800	Paid
503	103	22-SEP-26	Cash on Delivery	3200	Paid
504	104	23-SEP-26	Net Banking	1500	Failed
505	105	24-SEP-26	UPI	2750	Paid